Exact Kernel Improvements for SubexponentialSAT

A review of fifteen proposals, adaptive integral modes,
and measured improvements to packet coefficient extraction

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Abstract

We review the recent ProveIt proposals through commit `6d3adc45061535ebe6fa7cd3d9b37dff7dcfc22a`, including the eight newly placed second-round packets. One strong explicit compatible polynomial-space construction combines asymmetric packing with full-block Cauchy modes; the explicit normalized coefficient is approximately 66.6051, rather than the smallest individual displayed coefficient 76.9090. A compatible method using more memory has time coefficient 44.5376 and space coefficient 39.4959. These are comparison baselines, not new discoveries of this report, and neither is asserted globally optimal.

We obtain concrete improvements in two local interfaces. First, minimum-valuation elimination on the actual real-port crossing matrix gives an integral separation whose sorted weights dominate those of every integral rank-one representation of that same matrix. It therefore never increases the abstract weighted mode inventory. One checked sparse crossing block has 260 feasible tuples, against 3013 for the restricted, zero-pruned, valuation-normalized full-block Cauchy comparator. Second, we implement streamed Gray evaluation, anisotropic centered antipodal grids, and a projective roots-of-unity specialization for packet coefficient extraction over exact power-of-two quotient rings. A nineteen-case suite measures 2.16–7.86 times speedups for antipodal Gray evaluation against the shipped reference routine. On a quadratic-extension repeated-exponent case, projective evaluation reduces 19683 samples to 6561 and the working precision from 41 to 32 bits, with a measured 6.90 times speedup over that reference.

The elementary identities, Gray codes, Smith forms and permanent methods are classical. The contribution is their proved integration, precision accounting and implementation beyond this repository snapshot. The measured results are local-kernel timings. We do not prove a smaller unrestricted SAT logarithmic exponent, provide a complete fast SAT solver, or certify the full upstream ETH argument. Complete proofs of the new local interfaces, executable checks, raw measurements and immutable source identities accompany this report.

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1 Outcome and the exact scope of the improvement

The completed result is an exact, executable improvement to work that the reviewed proposals perform inside their contraction engine. It has two complementary effects: fewer abstract mode tuples for populated crossing blocks, and less work to evaluate a packet entry for each tuple. The underlying global counting reductions and recursive contraction identities remain explicit dependencies.

For fixed clause width k, write n for variables and m for clauses. The strongest justified density envelope in the reviewed corpus remains

$$\operatorname{poly}_k(n,m)\,2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}. \tag{1}$$

Here and below asymptotic expressions are for sufficiently large inputs. At fixed density $m \le Dn$ this has exponent $O_{k,D}(n/\sqrt{\log n})$; for unrestricted density it has exponent $O_k(n\log\log n/\sqrt{\log n})$. The $n\log\log n/\log n$ target in the precision-refinement packet depends on an unconstructed structured-mode summation interface. It is not an available solver whose runtime can be used as an experimentally tested baseline [1, 5, 6, 7].

Our generic coefficient algorithm improves dense grid evaluation from $O(RqG)$ to $O(RG)$ coefficient-ring operations, after reading the input. Here R is the number of affine factors, q the number of positive homogenized target exponents and G the product-grid size. Its antipodal version evaluates only $\lfloor G/2\rfloor$ points. A supported projective version evaluates $G/(1+\min b_j)$ points and removes factorial precision loss. All these algorithms stream their grids and retain polynomial space. The adaptive matrix algorithm has polynomial preprocessing cost and a pointwise non-increase theorem for the abstract mode inventory.

These statements do not turn the measurements into full SAT runtimes. The existing contraction proof charges packet entry work as a lower-order term. Improving that term alone leaves its leading logarithmic order unchanged. Dense fully populated crossing matrices also retain the known slope-four Smith profile, so the adaptive algorithm supplies no uniform replacement of that slope by a larger constant.

2 Review cutoff, proposal inventory and combined baseline

The review began at ProveIt commit `ab2fbc9e03d5c3d0ad6dec8686960bfe498b0b06`. At that point there were seven indexed packets and seven incoming archives. A closing check found one further matching-packing packet, followed by the placement commit `6d3adc45061535ebe6fa7cd3d9b37dff7dcfc22a`. We read that additional report and relevant code. The final inventory is fifteen packets. At the closing snapshot `docs/incoming/` contains its index and the research packets are under `Computability/SubexponentialSAT/Research/`.

The vendored preprint is pinned upstream to `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`. The repository's own trust statement says that placement did not verify the mathematical results or run a formal development. This report performs a scoped analytic and interface review; it does not replace that statement with a claim of full formal verification [1, 2, 3].

2.1 What each packet contributes

In the next table, coefficients refer to the normalized contraction scale

$$\operatorname{poly}(s+\mathcal{L})\,2^{(C+o(1))s/\sqrt{\log_2 s}}, \tag{2}$$

where s bounds root Boolean wires, initial precision and truncation order, the root degree is at most $\lceil s^{9/10}\rceil$, and $\mathcal{L}$ is the description length. They are not numerical coefficients on SAT variables.

Packet	Contribution retained in the comparison	Limit
Contraction improvements	Exact taper, leading precision $6MD/u$, coefficient 648.4411, streamed homogeneous finite differences.	No measured solver.
Factorial Cauchy modes	Integral modes with slope three.	Same logarithmic order.
Precision refinement	Exact recovery losses and real taper minimax.	Faster summation is conditional.
SAT density improvement	Uniform density envelope (1).	No better unrestricted order.
ETH density refinement	Related density bound and local odd-unit Smith profile.	Local profile alone is not all-level grouping.
Scalar parity refinement	Scalar lift and smaller incidence-degree cap.	New factorization still missing.
Asymptotic analysis	Restricted barriers and explicit missing interfaces.	No improved unrestricted algorithm.
Recursive refinement	133.2074; missing-residue Cauchy slope 64/21.	Some finite inventories worsen.
Packet refinement	95.6639; nearly balanced loads and additive precision guard.	Same order.
High moments	94.7482 via moment control; degree-sensitive specializations.	Setup cost and eventual thresholds matter.
Balanced packing	94.7482 via constructive two-stage balancing.	Same order.
Matching refinement	94.7482 via matching potentials and thin gaps.	Additional late packet; no new extraction routine.
Round-two refinement	84.0663; exact crossing fraction and two-stage alphabet.	No completed stronger-order interface.
Asymmetric balancing	76.9090; uses smaller fresh-side degree below the root.	Its displayed basis has slope three.
Joint Cauchy	All-level full crossing-block slope four. Individual coefficient 319.6858; cached/Strassen option 225.5971.	Its own schedule is weaker; cached option uses more memory.

The separate 49-page manuscript credited to Claude has the same relevant unrestricted logarithmic order and does not provide a directly comparable explicit coefficient on (2). Numerical domination of that independent algorithm is therefore not established here. The inherited OpenAI manuscript's Appendix D is the comparison target, rather than the weaker endpoint described by its archived Lean scope material.

2.2 Combine compatible ingredients before claiming a gain

Asymmetric packing supplies near-unit padding, at most

$$(2(1-1/r)+o(1))M/D$$

families, child size $(2/r+o(1))M$, and precision $(6+o(1))MD/u$. The joint Cauchy basis has the same source and target groups, the same family multiplicity, and weights

$$a+1-i+4j-2s_2(j).$$

It changes neither child wiring nor the fresh-side degree. Substituting it for the slope-three basis multiplies the leading entropy coefficient by $\sqrt{3}/2$ [8, 9].

For the explicit asymmetric schedule $r=16$,

$$u_0=\lfloor \lambda/81 \rfloor, \qquad u_d=\left\lfloor \frac{\lambda}{108\,4^{d-1}} \right\rfloor \quad (d\ge 1), \qquad \lambda=\lceil \log_2 s \rceil,$$

the resulting coefficient is

$$C_{\mathrm{poly}}=\frac{\pi\sqrt{15/8}}{\ln 2}(9+\sqrt{3})=66.6051434076\ldots. \tag{3}$$

The boundary version of its fixed-$r=16$ family approaches 66.1927236936. A further numerical optimization of a two-stage family gives approximately 66.12047546 with root alphabet 17 and tail alphabet 13. This last number is recorded as a schedule-model infimum, not as a new implementation theorem. None of these values is a lower bound or a global optimum over all schedules. Our practical comparison does not rely on advertising a modest schedule retuning as a structural breakthrough.

When more memory is allowed, the joint packet's exact-weight indexing, cached group contractions, and three-block Strassen construction also compose with this schedule. For $r=16$ split the groups into $5,5,6$. Set

$$K_0=\frac{9\pi\sqrt{2}}{\ln 2}, \quad A=\sqrt{30/256}, \quad B=\sqrt{150/256}, \quad J=\sqrt{120/256},$$

and, with $\omega=\log_2 7$,

$$S=\sqrt{\frac{170-[1-(\omega-2)^2]50}{256}}.$$

The directed inter-block pair counts are $50,60,60$. The first child scale is $K_1/K_0=\sqrt{3}/12$; subsequent scales have ratio $1/4$. In the recurrence, cached child evaluations contribute AK_d before the recursive cost, and table/matrix work contributes $\max(B,S)K_d$. Since $A+S\sqrt{3}/12<S$ and $A(1+\sqrt{3}/9)<S$, the root matrix work dominates. Thus one available combined comparison has

$$C_{\mathrm{time}}=K_0S=44.5375718550\ldots, \qquad C_{\mathrm{space}}=K_0J=39.4959495337\ldots. \tag{4}$$

This is a different space regime from (3). The numerical calculation and its assumptions are reproducible in the accompanying code.

3 Exact packet coefficient extraction

Let $f \in \mathbb{Z}[U]$ be monic and $\ell \geq 1$. Work over

$$\mathcal{A}_p = (\mathbb{Z}/2^p\mathbb{Z})[U,V]/(f(U), V^\ell).$$

For this section f may be reducible and the ring may contain nilpotents. Its integral version is free as a $\mathbb{Z}$-module in the usual monomial basis, so coordinatewise exact division after sufficient precision is valid.

Given R affine forms, the required entry is

$$c_\beta = [Z^\beta] \prod_{r=1}^{R} \left(a_{r0} + \sum_{j=1}^{q} a_{rj} Z_j \right) \pmod{2^p}. \tag{5}$$

An impossible target $|\beta| > R$ gives zero. Otherwise put

$$b = (R - |\beta|, \beta_1, \ldots, \beta_q), \qquad H(X) = \prod_{r=1}^{R} \sum_{j} a_{rj} X_j,$$

so that $|b| = R$ and $c_\beta = [X^b]H$. A coordinate with $b_j = 0$ can be set to zero and removed from all evaluation loops. Below, q denotes the number of remaining positive coordinates. Define

$$G = \prod_j (b_j + 1), \qquad \nu = \sum_j v_2(b_j!) = \sum_j (b_j - s_2(b_j)), \qquad O = \sum_{j:\, b_j \text{ odd}} b_j. \tag{6}$$

The shipped finite-difference implementation uses exactly G grid points and working precision $p + \nu$ [4]. It recomputes each of the R row values from every grid coordinate at every point. The inspected later proposals retain this routine or the older direct row expansion; none supplies the three implementations introduced here.

3.1 Gray traversal without any precision change

A reflected mixed-radix Gray traversal changes exactly one coordinate by $+1$ or -1 at each transition. If $L_r = \sum_j a_{rj}\gamma_j$ is the current row value, changing coordinate j by ε updates it by

$$L_r \longleftarrow L_r + \varepsilon a_{rj}.$$

Only R row updates are needed. The signed binomial weight can also be updated without recomputing its product. For an increasing coordinate,

$$w_{\text{new}} = w_{\text{old}} \frac{b_j - \gamma_j}{\gamma_j + 1},$$

and the decreasing formula is analogous. The division is an exact integer division of the entire positive weight, followed by a sign flip. It is not division by a possibly even residue in $\mathcal{A}_p$.

Proposition 3.1 (Streaming arithmetic cost)**.** *For $R > 0$, full-grid Gray evaluation computes (5) at the same $p+\nu$ working precision as the shipped reference, with G evaluations and $O(\text{input size} + RG)$ coefficient-ring operations. It stores the input, one grid point, R row values, one integer weight and one accumulator.*

Proof. Initialize all row values at the initial grid point. Each of the $G-1$ transitions uses R additions/scalings. At each point the product uses R ring multiplications, followed by one weighted accumulation. The standard finite-difference identity is unchanged. The integer weight has at most $R+1$ bits, because the product of binomial coefficients is at most 2^R. The mixed-radix counter can scan $O(q)$ positions on one transition, but its aggregate work over a grid with all radices at least two is $O(G)$, by the geometric bound on carries. No worst-case constant delay assertion is needed. □

3.2 Anisotropic centered grids and exact antipodal pairing

Choose the smallest positive integer step that centers the grid integrally:

$$h_j = \begin{cases} 1, & b_j \text{ even}, \\ 2, & b_j \text{ odd}, \end{cases} \qquad x_j(\gamma_j) = h_j\gamma_j - h_j b_j/2. \tag{7}$$

Define $w(\gamma) = (-1)^{R-|\gamma|}\prod_j \binom{b_j}{\gamma_j}$.

Theorem 3.2 (Anisotropic antipodal extraction)**.** *Assume $R > 0$. Let $\mathcal{P}$ contain one representative of each noncentral pair $\{\gamma, b-\gamma\}$ in the product grid. Then*

$$\sum_{\gamma\in\mathcal{P}} w(\gamma)H(x(\gamma)) = \left(2^{O-1}\prod_j b_j!\right)[X^b]H. \tag{8}$$

The parenthesized integer has valuation $\nu + O - 1 \geq 0$. Consequently the target coefficient modulo 2^p can be recovered from exactly $\lfloor G/2 \rfloor$ evaluations at working precision

$$P = p + \nu + O - 1. \tag{9}$$

Proof. The product of the b_j-th finite-difference operators annihilates every monomial having exponent below b_j in at least one coordinate. Since H is homogeneous of total degree $|b|$, the only surviving monomial is X^b. Its product difference equals $\prod_j b_j! h_j^{b_j}$. Thus the full-grid sum equals $2^O \prod_j b_j! [X^b]H$.

Complementation sends x to $-x$. It changes the signed binomial weight by $(-1)^R$ and the value of H by the same factor. The two contributions therefore agree. An unpaired center exists only when all b_j are even; there $x = 0$ and $H(0) = 0$ for $R > 0$. Pairing proves (8) as an identity over the integers. If $O > 0$ then $O \geq 1$. If $O = 0$, at least one positive exponent is even, so $\nu \geq 1$. The denominator is consequently integral in both cases.

Fix the original coefficient representatives at creation precision p. Compute the representative sum modulo 2^P. Its coordinates are divisible by $2^{\nu+O-1}$. Divide those canonical integer coordinates by this power of two and multiply by the inverse of the odd part of $\prod_j b_j!$ modulo 2^p. Errors before division are multiples of 2^P, hence errors after division are multiples of 2^p. Freeness of the integral quotient algebra justifies the coordinatewise operation. No even element is inverted. □

Three cases make the precision effect transparent:

Target pattern	Evaluations	Working precision
All positive b_j even	$(G-1)/2$	$p + \nu - 1$
Exactly one odd exponent, equal to 1	$G/2$	$p + \nu$
All positive $b_j = 1$	2^{R-1}	$p + R - 1$

The last line is the classical Glynn permanent specialization. The squarefree case pays more precision even though it uses half as many points. The all-even case uses one fewer bit than the old full grid.

Proposition 3.3 (Optimal integer centered step valuation)**.** *Among symmetric arithmetic grids with integer coordinates and lengths $b_j + 1$, the choice (7) minimizes the two-adic valuation of the difference denominator. The loss $\nu + O - 1$ is sharp for this paired functional.*

Proof. An integer-centered grid requires $h_j b_j$ even. An odd b_j forces $v_2(h_j) \geq 1$; an even b_j allows $h_j = 1$. This gives the minimum $\sum_j b_j v_2(h_j) = O$. For $H = X^b$ the coefficient is one and the representative numerator equals the displayed denominator, proving the sharpness statement for this functional. □

3.3 Streaming the half-grid

If an exponent b_j is odd, restrict that coordinate to $0 \leq \gamma_j \leq (b_j - 1)/2$ and leave all others unrestricted. This is one rectangular set containing exactly one point of each pair. If all exponents are even, use a disjoint union of boxes: earlier coordinates are fixed at their midpoints, the first noncentral coordinate j lies strictly below its midpoint, and subsequent coordinates are unrestricted. This covers $(G - 1)/2$ representatives and omits only the zero center.

Each box admits reflected Gray traversal. Row values need initialization at most q times. The total initialization cost is $O(Rq^2)$, absorbed by $O(RG)$ because an all-even positive profile has $G \geq 3^q$. Within a box each transition updates row r by $\pm h_j a_{rj}$. The storage remains polynomial. Degree zero and impossible targets are handled directly, outside the paired formula.

4 Projective roots-of-unity extraction

The half-grid is not always the smallest useful grid. Homogeneity can remove a whole coordinate, provided the other coordinate filters have the required orthogonality. This is related to classical generalized permanent identities and monomial Waring decompositions [10, 11, 12]. We give the exact modular form needed here.

Theorem 4.1 (Projective coefficient filter)**.** *Let H be homogeneous of degree $R = \sum_j b_j$, with all retained $b_j > 0$. Choose k with $b_k = \min_j b_j$. Put $d_j = b_j + 1$. For every $j \neq k$, suppose a ring element ω_j satisfies*

$$\omega_j^{d_j} = 1, \qquad \sum_{t=0}^{d_j - 1} \omega_j^{tm} = \begin{cases} d_j, & d_j \mid m, \\ 0, & d_j \nmid m. \end{cases} \tag{10}$$

The identities may hold exactly or modulo a chosen working power of two. With $X_k = 1$ and $X_j = \omega_j^{t_j}$ for $j \neq k$,

$$\sum_{\substack{0 \leq t_j < d_j \\ j \neq k}} \left(\prod_{j \neq k} \omega_j^{-b_j t_j} \right) H(X) = \left(\prod_{j \neq k} d_j \right) [X^b] H. \tag{11}$$

There are exactly $G/(1 + \min b_j)$ evaluations. To recover modulo 2^p, sufficient working precision is

$$P = p + \sum_{j \neq k} v_2(d_j). \tag{12}$$

Proof. Expand H. Orthogonality retains a monomial X^α only when $\alpha_j \equiv b_j \pmod{d_j}$ for $j \neq k$. Since $0 \leq b_j < d_j$ and $\alpha_j \geq 0$, this means $\alpha_j = b_j + t'_j d_j$ with $t'_j \geq 0$. Homogeneity gives $\alpha_k = b_k - \sum_{j\neq k} t'_j d_j$. Every $d_j = b_j + 1$ exceeds b_k. Thus any positive t'_j would force $\alpha_k < 0$, and the only surviving monomial is X^b. Its contribution is $\prod_{j\neq k} d_j$ times its coefficient. If the identities hold only modulo 2^P, the same proof is a congruence modulo 2^P. Divide canonical integer coordinates by the explicit power of two in the denominator and invert its odd part. The precision loss is exactly the sum in (12). □

Choosing a minimum exponent is essential. For example, fixing the exponent-two coordinate in target $X_0^2 X_1$ and sampling X_1 only at ± 1 aliases the monomial X_1^3. Mere multiplicative order is also insufficient over an arbitrary ring; the required assertion is the orthogonality in (10).

4.1 Implemented specialization: exponents one and two

For $b_j = 1$ use roots $1, -1$. For $b_j = 2$ use $1, \omega, \omega^2$ with $\omega^2 + \omega + 1 = 0$. These imply the needed orthogonality in every commutative ring. The code verifies the cubic relation at the working precision. For the existing ring defined by $f(U) = U^2 + U + 1$, the element $\omega = U$ supplies the root directly, including with a truncated V axis. No extra extension dimension is allocated implicitly.

If there are q_1 exponents equal to one and q_2 equal to two, then:

Pattern	Evaluations	Working precision
$q_1 > 0$	$2^{q_1-1}3^{q_2}$	$p + q_1 - 1$
$q_1 = 0$	3^{q_2-1}	p

In the mixed case the evaluation count matches the paired grid, while the precision saves q_2 bits. In the all-two case it saves almost another factor $3/2$ in samples and removes all factorial precision loss.

The source contraction uses an unramified residue extension of degree $e = 4\lceil \log_2 s \rceil$. Since e is even, its residue field contains the nontrivial cubic roots. A chosen residue root lifts modulo 2^P because the derivative $2\omega + 1$ is a unit. This is a finite-precision Hensel statement; it does not assert that the uncompleted rational coefficient field already contains an exact complex cube root. A caller using a different defining polynomial supplies a correctly lifted root. The provided general theorem also describes other supported odd root orders; the executable specialization deliberately restricts positive exponents to one and two.

Reflected Gray traversal of the root-index grid again changes one coordinate. Precompute each adjacent root difference times each row coefficient. Then one transition uses R additions; the character weight updates by a unit root multiplication. The loop has $O(RG/(1 + \min b_j))$ ring operations, plus polynomial initialization, and polynomial storage.

5 Adaptive integral modes on actual crossing blocks

The other improvement changes which modes exist before any packet-entry coefficient is requested. The full-block basis uses a convenient ordering of padded slots. Restricting it to real ports leaves cancellations and linear dependencies that pointwise source/target valuation normalization does not remove.

For this section the kernel coefficients belong to the unramified chain ring

$$R_p = (\mathbb{Z}/2^p\mathbb{Z})[U]/(f(U)),$$

where $f \bmod 2$ is irreducible. The two-adic valuation of a nonzero element is the minimum coordinate valuation; zero has capped valuation p. An element of valuation zero is a unit. The kernel matrix itself contains no nilpotent V dependence. The surrounding tensor entries may still belong to $R_p[V]/(V^\ell)$.

5.1 Minimum-valuation elimination

Let K be the $m \times n$ matrix of one actual crossing orientation, restricted to its real row and column slots. At a residual matrix E, choose an entry E_{ij} of minimum valuation $e < p$. Write $E_{ij} = 2^e u$ with u a unit at precision $p - e$. Define

$$c_h = E_{hj}, \qquad r_k = (E_{ik}/2^e)u^{-1} \pmod{2^{p-e}}.$$

Use canonical lifts of the r_k to R_p and subtract cr^T. The pivot row and column vanish modulo 2^p. Repeating gives

$$K = \sum_{h=1}^{t} c_h r_h^\mathsf{T} \pmod{2^p}, \qquad v_2(c_h) \geq e_h, \quad e_1 \leq \cdots \leq e_t < p. \tag{13}$$

The extra precision needed to prepare a Cauchy kernel is a separate matter; the elimination itself never requests more than the matrix's p bits.

Theorem 5.1 (Exact adaptive separation and its cost)**.** *The algorithm above produces the nonzero Smith valuations of K and an integral rank-one separation with those weights. It takes $O(mn\min(m,n))$ ring operations, in addition to polynomial-time unit inversions, and $O(mn)$ stored ring elements.*

Proof. Every residual entry is divisible by 2^e. The quotient pivot is a unit, so the displayed row is integral at precision $p - e$. Since the entire pivot column is divisible by 2^e, ambiguity in lifting that row changes its outer product only by a multiple of 2^p. Direct substitution proves that the pivot row and column become zero.

Equivalently, swap the pivot to the top-left position, subtract integral multiples of its row and column and normalize the odd unit. These are invertible row and column operations over the chain ring. The result is a diagonal entry 2^e and a smaller residual matrix. Every residual entry remains divisible by 2^e, so subsequent pivot valuations cannot decrease. Induction therefore gives the Smith diagonal and the exact decomposition. There are at most $\min(m,n)$ pivots, each requiring a scan and a rank-one update of $O(mn)$ entries. The sum of stored source and target vector lengths is at most $\min(m,n)(m+n) = O(mn)$, in addition to the matrix. □

5.2 A pointwise comparison against every integral separated basis

Theorem 5.2 (Weight-profile domination)**.** *Suppose another integral representation of this same matrix is*

$$K = \sum_{j=1}^{N} 2^{w_j} a_j b_j^\mathsf{T} \pmod{2^p}, \qquad 0 \leq w_1 \leq \cdots \leq w_N < p.$$

Let $e_1 \leq \cdots \leq e_t$ be the adaptive weights. For every $1 \leq q \leq p$,

$$\#\{h : e_h < q\} \leq \#\{j : w_j < q\}. \tag{14}$$

In particular $t \le N$ and $e_h \ge w_h$ for $1 \le h \le t$. If all weights are positive, then for any budget $1 \le P \le p$,

$$\#\{\beta \in \mathbb{Z}^t_{\ge 0} : \sum_h e_h \beta_h < P\} \le \#\{\beta \in \mathbb{Z}^N_{\ge 0} : \sum_j w_j \beta_j < P\}. \tag{15}$$

Proof. Reduce the matrix modulo 2^q. Its Smith image is a direct sum of exactly $t_q = \#\{h : e_h < q\}$ nonzero cyclic chain-ring modules. Thus its minimal number of generators is t_q. In the alternative representation all terms with $w_j \ge q$ vanish. The image is contained in the submodule generated by the $N_q = \#\{j : w_j < q\}$ surviving target columns.

For completeness, submodule containment does bound the number of generators here. A finite module over a chain ring is a direct sum of cyclic modules. Its minimum generator number equals the dimension over the residue field of its socle, the submodule annihilated by 2. Inclusion of modules gives inclusion of their socles. Hence a submodule of a module generated by N_q elements needs at most N_q generators. This proves (14). Taking $q = p$ proves $t \le N$; if $e_h < w_h$, taking $q = e_h + 1$ contradicts the cumulative inequality. Finally map a feasible adaptive exponent tuple to its first t coordinates in the alternative system and append zeros. The total alternative weight is no larger, proving the injection in (15). □

This is a theorem about separated representations of a fixed matrix. It does not claim optimality of every way of contracting a tensor network, or of nonlinear cancellation among full mode tuples. It is stronger than a comparison only with untrimmed padded modes: it still applies after restricting an earlier basis to real ports, deleting zero terms and transferring every common source valuation into the target coefficient.

The matrix row and column sets lie within one source/target orientation. Every new source form therefore depends on the same source group's bits, and every new target coefficient belongs to the same target group. Freeze the generated source and target vectors as canonical representatives at their creation precision; later recursive precision increases reuse these same representatives. The exact factorization used in the source recursion survives. The degree, wire and packing invariants also survive. Polynomial setup cost and polynomial matrix storage are admissible because packet length is a fixed power of the root size in the supplied schedules.

5.3 A closed-form family with strictly stronger slopes

The preceding algorithm has a transparent special case. Fix a valuation level i, and let real row and column slots be $x_0 + Hj$, $0 \le j < N$, with $v_2(H) = i + h$, $h \ge 1$. Write $T = 2^i\tau$ with τ a unit and $H = 2^{i+h}v$ with v odd. The crossing matrix is

$$K_{jk} = \frac{\Gamma(T)/2^i}{\tau + 2^h v(j-k)}, \qquad \Gamma(T) = \mathcal{C}(T)/\mathcal{C}'(0).$$

Let $\delta = v_2(\Gamma(T)/2^i)$; for unit ζ, including the extension-field values used in the examples, $\delta = a+1-i$.

Corollary 5.3 (Dyadically spaced real ports)**.** *The nonzero Smith weights, truncated at precision p, are*

$$e_j = \delta + 2hj + 2v_2(j!) = \delta + (2h+2)j - 2s_2(j), \qquad 0 \le j < N. \tag{16}$$

Proof. Apply Cauchy elimination in the actual consecutive real-slot order. All denominators are units after division by 2^i. The target factor after j pivots contains $\prod_{k<j} 2^h v(j-k)$, of valuation $hj + v_2(j!)$, and so does the source factor. For a later row index $l \ge j$, $\prod_{k<j}(l-k) = j!\binom{l}{j}$ is divisible

by $j!$, with equality of valuations at $l = j$; entries before the diagonal vanish. The same argument applies on the column side. Dividing these known powers leaves triangular transformations with unit diagonals. The pivot therefore has valuation (16); the normalized triangular matrices are invertible. Equivalently, the Cauchy determinant formula and Vandermonde products show that consecutive principal minors attain the minimum determinant valuations at every order. □

The full padded profile corresponds to $h = 1$ and slope four. Slots spaced an additional factor 2^{h-1} have slope $2h + 2$ on this block. That is a real structural gain for the stated support pattern. General SAT encodings are not proved to generate this pattern uniformly, so it does not change (1).

6 Executed evidence: correctness and measured performance

The benchmark programs use Python 3.12.14 on the provided Linux x86-64 execution environment. They use only the standard library, fixed seeds, three timed repetitions, and rotated method order. Each timing fixture is checked against direct truncated polynomial multiplication outside the timed region. The full coefficients, seeds, ring specifications, precisions, raw repetitions and medians are included in the JSON data. The cases are deterministically generated packet-kernel inputs, not a sample of SAT formulas or a representative industrial benchmark suite.

6.1 Coefficient extraction measurements

Case	R	Old	Gray	Antipodal	Gain
Squarefree	14	1.815878	0.509625	0.231138	7.86
All exponents two	18	2.371582	0.759171	0.407105	5.83
Three exponents sixteen	48	0.970922	0.462456	0.278199	3.49
One odd exponent	13	0.095707	0.038842	0.019302	4.96
Quadratic extension	10	0.071759	0.034595	0.014858	4.83
Nilpotent extension	8	0.021168	0.012678	0.006339	3.34

Table 2: Selected packet extraction medians in seconds. Gain is the old runtime divided by the antipodal runtime; it is not a SAT speedup.

Across the nineteen cases in the general extraction suite, the median of the per-case speedup ratios for antipodal Gray evaluation is 4.56, with observed range 2.16–7.86. The column labelled "Gray" retains the old working precision and shows the benefit of incremental row evaluation separately from symmetry.

Case	R	Old	Antipodal	Projective	Gain
All two, small	6	0.000980	0.000333	0.000345	2.84
All two, quadratic	18	2.914805	0.553943	0.422439	6.90
All two, nilpotent	12	0.080007	0.024649	0.017121	4.67
All two, $p = 1$	12	0.044650	0.009691	0.006115	7.30
Mixed one/two	10	0.018711	0.005157	0.005131	3.65

Table 3: Projective specialization, median seconds. The same extension ring and same coefficient input are used for all methods within each row. Gain compares old and projective runtimes.

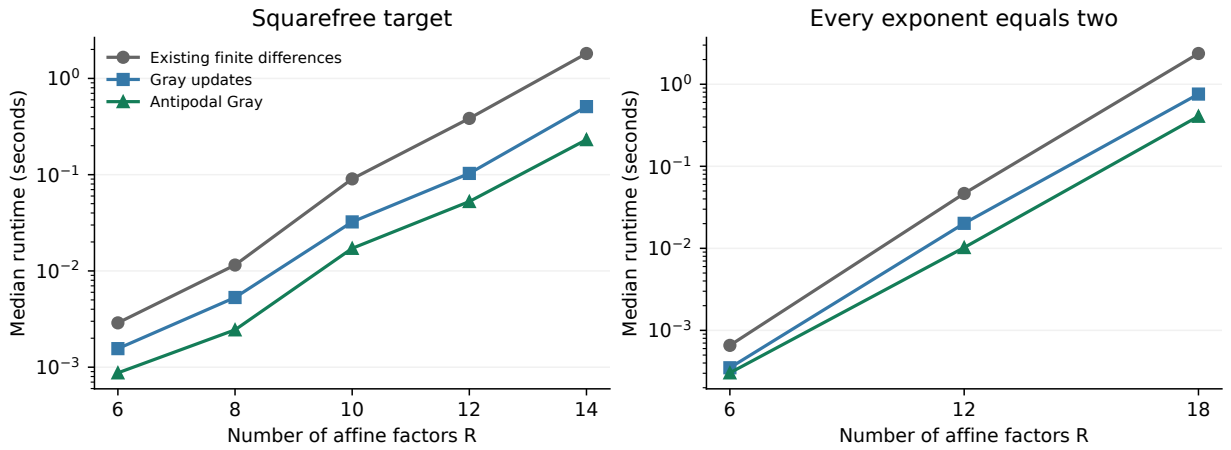


Figure 1: Measured scalar packet-entry runtimes, medians of three runs. Both panels are logarithmic in time. These are local coefficient extractions, not SAT solve times. The paired-exponent method also uses one fewer working bit than the old full grid.

The projective suite contains ten cases. Its all-two profiles improve on the antipodal kernel by approximately 1.31–1.58 times in the larger tested cases. Mixed one/two profiles have the same sample count and are approximately tied; the tiny three-axis all-two case is slightly slower than antipodal evaluation. The data therefore support a dispatchable specialization, not a claim that it is universally fastest. All comparisons use the same coefficient ring in each row. We do not compare a scalar-ring method against a method silently using a larger extension.

6.2 Adaptive mode inventories

L	i	p	m	Old modes	New modes	Old tuples	New tuples
32	0	24	8	5	4	13	11
32	0	40	8	9	7	64	41
64	0	48	16	11	8	98	58
64	0	64	8	15	8	240	87
64	2	40	16	31	16	2,862	1,023
64	2	48	8	29	8	3,013	260

Table 4: Exact local inventories over the quadratic extension with interpolation value $\zeta = U$. There are m real rows and columns. Mode and tuple counts compare the normalized restricted padded basis with adaptive separation.

For the last row, the comparator has 29 surviving normalized modes and the adaptive method has eight. The comparator is the full joint Cauchy basis restricted to the selected real slots, with zero modes removed and all common source valuations moved into target coefficients. Thus the gain is not obtained merely by counting padded slots as active. The counts use the strict budget $\sum w_j \beta_j < p$. They measure abstract feasible tuples, not the number of nonzero complete tensor terms and not a wall-clock speedup. Polynomial factorization setup must also be paid in an implementation.

6.3 Correctness checks and independent review

The delivered checks include the following independent comparisons.

- Gray and antipodal extraction: 1392 exhaustive small scalar cases, 900 randomized cases across five rings, and 340 reflected-grid coverage checks. The ring fixtures include reducible monic quotients and nilpotents.
- A second extractor audit: 741 target profiles and 20860 antipodal orbits, 280 integer-oracle comparisons and 280 canonical-representative invariance comparisons, plus all-even precision fixtures.
- Projective extraction: 10028 exhaustive cases and 813 randomized comparisons among direct multiplication, the shipped routine, antipodal Gray and projective extraction. Rejection tests include a cubic root valid modulo two but invalid at the required augmented precision, and a nonminimal-pivot alias example.
- Adaptive separation: 720 arbitrary matrices, 6840 independent determinant-minor checks, 720 actual crossing matrices and 180 full padded-profile checks. All decompositions reconstruct their input modulo the specified power of two.

The new checks were also run under optimized Python. Their validation gates use explicit exceptions or unconditional comparisons, rather than depending on removable assertions. These checks support the implementations and catch precision errors; the all-size theorems are established by the proofs above. No Lean or Rocq certificate is claimed.

7 Integration and reproducibility

The archive is a self-contained local implementation packet. It includes the report, its complete LaTeX source and figures, new kernels, the unmodified baseline coefficient routine, mathematical checks, raw benchmark records, source fingerprints and review notes. No remote repository was modified while preparing it.

7.1 Preserve the source interface

An integration should proceed through the existing exact interfaces:

1. Keep the best existing reduction, packet labels, precision taper, creation-time representatives and child graph construction.
2. For each crossing orientation at its requested precision, form the matrix only on actual real ports. Optionally replace its fixed separated basis with `adaptive_modes.separate`. Keep source and target group support unchanged. Retain a fixed-basis path when its setup cost is preferable.
3. When a packet tensor requests a coefficient of its product of affine row factors, pass the same canonical input entries to the Gray, antipodal or supported projective extractor. Return the answer at the packet's creation precision. Temporary extra precision must not change that creation precision or the immutable description.
4. Use projective extraction only when every positive homogenized exponent is supported and the required root is validated at working precision. Otherwise use a general extractor. Plain Gray

evaluation is available when extra bit precision is undesirable.

Nothing in these replacements assumes that canonical lifting preserves addition or multiplication. They evaluate the polynomial in the already fixed representatives and then return its residue. Each mode tensor remains a valid integral packet description. The adaptive comparator theorem continues to hold at every requested cutoff, so it composes with both depth-first polynomial-space enumeration and the existing cached mode-summation option, while retaining their respective resource regimes.

7.2 Commands

From the extracted archive root, the following commands require Python 3 and its standard library:

```
python3 code/run_checks.py
python3 code/verify_and_benchmark.py --output replay/benchmarks
python3 code/benchmark_projective.py
python3 code/check_certificates.py
```

The benchmark commands are optional and regenerate timing data; exact check outputs are supplied so that reproduction is not required to read the report. Consult `README.md` for the final command-line switches and paths. The PDF can be rebuilt with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -outdir=build report.tex
```

The figure and generated table sources are included. No network access or upstream script execution is required for these checks.

8 Research conclusion and remaining barrier

The review changes what a defensible comparison should mean. The smallest individual certificate is weaker than the compatible combination of its packing schedule with another packet's Cauchy basis. The more-memory combination is stronger again in time but does not preserve polynomial space. None of these upper-bound comparisons measures a SAT runtime ratio.

The useful new result here is a set of exact replacements with proved local effects and measured kernel behavior. Adaptive separation is optimal for the cumulative integral rank profile of a fixed crossing matrix. Anisotropic pairing halves the grid, with exact precision accounting that can save a bit rather than spend more bits. Projective root filters remove a complete coordinate and factorial loss when the available roots and target multiplicities permit it. Incremental row updates provide a separate factor-q arithmetic improvement without increasing precision.

To change the unrestricted exponent in (1), one still needs a theorem controlling the global mode sum or a smaller valid representation of the dense counting reduction. Local Smith optimality also clarifies the limit: another integral basis of a dense fixed block cannot improve its weight profile. A further breakthrough must exploit structure across blocks or groups, change the representation, or sum contributions without paying for every feasible tuple. The code and proofs here provide concrete, compatible improvements while leaving that open problem stated precisely.

A Why the classical identities are credited

In the squarefree case, homogeneous coefficient extraction is a permanent, and the antipodal formula is Glynn's identity. Mixed-radix Gray traversal and generalized formulas for repeated rows or columns are established techniques [10, 11]. The projective grid size also appears in monomial Waring decompositions [12]. Smith normal form and Cauchy elimination are classical algebra. We make no claim of inventing these methods or of global research priority.

The difference from the reviewed source corpus is specific: it lacked these implemented packet-entry algorithms and their power-of-two precision rules, and it used predetermined Cauchy mode systems rather than precision-aware elimination of the actual real-port matrix. Theorems 3.2, 4.1, and 5.2 state the exact interfaces and limits of that contribution.

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